Sgidi Mamolulo Jeminah

47
68/
65
69
94%
=94%

Module #	671101000-KM-06:
NQF Level	level 4,
Notional hours	130
Credit(s)	Cr 13
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

ROTATING ELECTRICAL MACHINERY
FORMATIVE ASSESSMENT

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About the Formative Assessment	
Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room.
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ **points.** In order to meet the pass mark, learners are required to achieve a minimum of **80%** of the total marks.

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Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: Sasa Mamoloko	Manager:
Trainer:	Mentor / Coach:
Facilitator	
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner: Sasa Mamoloko $Sasa M·G

Learners Name: Sasa Mamoloko

Learner's signature: $Sasa M·G Date:

Facilitator Name:

Facilitator's signature: Date:

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ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____

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Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: Sasa Mamololo	Signature M. g
Facilitator Name:	Signature:

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LEARNING UNIT 1:

KM-06-KT01: Rotating electrical machinery - AC motors (35%)

Learning Outcome 1:	KT0101 Construction of alternating current motors
Learning Outcome 2:	KT0102 Principle of operation of alternating current motors
Learning Outcome 3:	KT0103 Configuration of motor connections
Learning Outcome 4:	KT0104 Types of single phase and three phase alternating current motors
Learning Outcome 5:	KT0105 Application of alternating current motors
Learning Outcome 6:	KT0106 Testing principles of single phase and three phase alternating current motors
Learning Outcome 7:	KT0107 Protection of motors
Learning Outcome 8:	KT0108 Calculation of motor properties

Task	Questions Description	Marks
2	a) Explain the principle of operation of alternating current motors	4
	b) Describe and explain how tests are conducted on alternating current motors	4
	c) Describe and explain protection devices used for alternating current motors	5
	d) Compare the advantages and disadvantages of single- and three-phase motors	6

@ An Ac motor operates on the principle of electromagnetic induction. When Alternating current flows through the Stator windings, it produces a rotating magnetic field. This rotating magnetic field induces a current in the motor, which create it's own magnetic Field

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LEARNING UNIT 2:

KM-06-KT02: Rotating Electrical Machinery - DC motors (35%)

Learning Outcome 1:	KT0201	Construction of direct current motors
Learning Outcome 2:	KT0202	Principles of operation for direct current motors
Learning Outcome 3:	KT0203	Configuration of direct current motor connections
Learning Outcome 4:	KT0204	Types and application of direct current motors
Learning Outcome 5:	KT0205	Testing principles of direct current motors
Learning Outcome 6:	KT0206	Protection of direct current motors
Learning Outcome 7:	KT0207	Calculation of direct current motor properties

Task	Questions Description	Marks
2	a) List the different types of direct current motors and describe their construction	2
	b) Describe the applications of different types of direct current motors	2
	c) Describe with the aid of diagrams, the configuration of motor connections, power circuits and control circuits.	2
	d) Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation, short circuits.	2
	e) Calculate direct current motor properties	2

a) b) Shunt wound DC Motor - field windings connected in parallel to armature. And has a stator.

c) Compound wound DC Motor - combination of series and shunt windings.

- Offers high starting torque and fairly constant speed.

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(d) ① Current rating
- Set slightly above normal motor current.
- Protect against overload and locked rotor.

② Time delay
- Short delay - trips during starting
- Correct delay - allows start-up, trips on fault.

③ Protection in different conditions

④ Locked rotor
- Prevent burning of windings
- Current very high

⑤ Overload During operation
- Prevents overheating
- Currently slightly above normal
- Relay trips after longer time.

(d) ⑥ Back EMF Equation.
- Power of motor ($P = V \times I$)
- Mechanical power developed - ($Pm = Eb \times Ia$)
- Torque equation - $T \propto \phi I_a$

For example calculation. Given - $V = 200\,V$
$I_a = 10\,A$
$R_a = 1\,\Omega$

$Eb = 220 - (10 \times 1) = 210\,V$

Mechanical power

$Pm = 210 \times 10 = 2100\,W$

(2)

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① Generator (DC Generator)
- Converts mechanical energy to DC electrical energy.
- A coil rotates in a magnetic field
- Output is direct current using a commutator.

② Alternator (AC Generator)
- Converts mechanical energy to AC electrical energy
- Uses slip rings instead of commutator
- produces Alternating current (AC)

(b) ① Alternators
- Power stations (electricity generation)
- Car charging systems (Automotive alternators)
- Diesel generators.

② DC Generators
- Battery Charging
- Electroplating
- Welding machines

(c) ① Overload protection
- protects against excess current
- prevents overheating

② Short circuit protection
- Trips instantly during fault
- prevents serious damage.

③ Earth fault protection

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SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

OUTCOMES		1	2	3
1	KT0301 Construction of alternators and generators			✓
2	KT0302 Principle of operation of alternators and generators			✓
3	KT0303 Configuration of alternator and generator connections			✓
4	KT0304 Types and application of alternators and generators			✓
5	KT0305 Testing principles of alternators and generators			✓
6	KT0306 Protection of alternators and generators			✓
7	KT0307 Calculate properties of alternators and generators			✓

Sejsa M.g — 18 March 2026

Learner Signature **Date**

Facilitators' Signature **Date**

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DECLARATION BY LEARNER			
I, Sasa Mamoloko Germiah .....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
Sasa MG			
Learner	**Date**	**Facilitator**	**Date**

DECLARATION BY THE ASSESSOR			
I ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	**Date**	**Moderator**	**Date**

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Notification of Assessment

Assessment will take place as follows:

	Theory	**Practical**
Date		
Time	9H00	9H00
Venue	Employer premises	Employer premises
Instruments	Written assignment	Practical Assessment

Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

ETDP PRACTITIONER

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5. Complete learner toolkit

6. Complete assessor toolkit

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